

Seung Hoon Lee / Curriculum Vitae

nalee0015@gmail.com / +82-10-9899-1786

Education

Hanyang University, School of Architecture, Seoul, South Korea

Mar. 2016 - Feb. 2022

Bachelor of Architecture, GPA 3.86/4.5, Graduated Cum Laude

- **Major Honors:** 2nd winner at Graduation Exhibition, President of Architecture Student Council(2018), Recipient of Merit-based Scholarship(2019)
- **Thesis:** Proposed a computational form-generation methodology by integrating structural analysis (Karamba 3D) and genetic algorithms (Galapagos), while developing a hybrid joinery system utilizing standardized beams and custom 3D-printed joints to optimize construction feasibility for challenging environmental constraints.
- **Focus Area:** Nature-informed Generative design, Structural Analysis and Optimization, 3D Printed Joinery System

Work Experience

JK-AR, Architectural Office, Seoul, South Korea

Jul. 2022 - present

Architectural designer, Algorithm designer / Full time

- **Isotropic Tree:** Developed a form-finding algorithm including structural analysis for extracting principal stress lines and visualizing algorithms for renders, motion diagrams, and construction diagrams as Project Manager.
- **Pavilion of Contemplation:** Oversaw the full project lifecycle as Project Manager, from schematic design to on-site construction, while meticulously managing technical constraints, including automated generation of dog bone details for CNC-fabrication and prefabrication strategies, to facilitate the successful dismantling and reassembly of the structure across two distinct sites.
- **Tree VI:** Directed the comprehensive workflow as a Project Manager, from developing a reciprocal structure generation algorithm on a universal surface to on-site construction, proposing curved joinery shape to ensure an expedited fabrication and assembly process.
- **The Nest:** Established foundational geometric rules for the design algorithm and implemented an automated documentation system as Project Manager, while orchestrating multidisciplinary collaboration with structural and lighting engineering firms.
- **Keywords:** Nature-informed Computational Design, Reinterpreting Historical Architecture, Automated Documentation System

Projects & Activities

Jul. 2024 - present

Code of Phenomenon, Representing Natural Patterns with Generative Algorithm

Researcher, Algorithm Designer

- **Generative Pipeline Development:** Developed a parametric, algorithm-driven pipeline for AI-generated video inspired by natural pattern systems, utilizing Grasshopper to simulate motion paths and generate multi-layered control maps, including outline, alpha, depth, and light maps, for Stable Diffusion workflows. Investigated the perceptual thresholds of AI-generated complexity to identify levels recognized as natural patterns. Produced images and videos as part of the "Noncircle" series.
- **Neural Heritage, Asking about Modern Enlightenment:** Awarded 2nd Place at the AI Heritage Competition (Gongju City Government) for exploring new visual languages through generative AI. The project implemented ControlNet and IP-Adapter frameworks to transfer natural patterns onto historical artifacts, investigating their potential application to architectural surfaces.
- **Stained Glass for the Anthropocene:** Received an Honorable Mention at the "Seoul Is Museum" Public Art Artist Discovery Competition for investigating the application of AI-generated natural patterns into architectural elements. Proposed strategies to mitigate urban environmental issues by translating generative digital patterns into functional architectural strategies for human-non-human coexistence.

2023 Seoul Biennale of Architecture and Urbanism (2023 SBAU)

Sep. 2023 - Nov. 2023

Team Leader, Participated Artist

- Selected through the Seoul 100-Year Masterplan Exhibition Open Call (\$3,500 prize), and exhibited work at the 2023 Seoul Biennale of Architecture and Urbanism.

- Proposed an “artificial succession” strategy, the natural developmental stages of a forest, to transform the historic Seun buildings into vertical park platforms by integrating the natural context of the Cheong-gye Stream through three distinct architectural and mechanical intervention machines.
- Developed algorithmically generated systems designed to regulate light, rainwater, and soil, to facilitate large-scale ecological restoration within the aged structure of the site.

Architect Resources Development Project

Dec. 2022 - Apr. 2023

Team Co-Leader, Head Designer

- Secured a \$15,000 research grant to support international architectural initiatives for emerging designers, leading to a nominated project for LAGI 2022 Mannheim in collaboration with BUGA23.
- Inspired by the image of a weeping willow, the main structure, composed of stacked timber elements and Triboelectric Nanogenerator (TENG) modules, expresses the dynamic movement of wind. The entire structural system and its embedded modules are fully controlled through a computational workflow developed in-house, enabling the harvesting of wind energy through an artistic architectural form.
- Developed a custom wind-motion simulator from scratch, utilizing directional stochastic algorithms to visualize the randomness of wind motion.
- Exhibited the final outcomes at the Land, Infrastructure, and Transport Technology (LITT) Fair 2023 at COEX, Seoul

2019 Seoul Biennale of Architecture and Urbanism (2019 SBAU)

Sep. 2019 - Nov. 2019

Project Team Co-Leader

- Nominated as part of the exhibition team following a 2nd Place award at the 2019 UAUS Pavilion Exhibition.
- Developed a computational design system to parameterize and coordinate hybrid tetrahedral and cubic geometric systems, comprising over 1,300 unique joints and structural members.
- Materialized geometrically complex digital models into prefabricated components and managed the on-site assembly process to ensure structural precision and geometric integrity.

Pavilion Exhibition at Union of Architecture University Students (UAUS)

Mar. - Jun. 2018, 2019

Student Representative of Hanyang University (2018), Team Co-Leader (2019)

- 2018 (2nd Place Winner): Delivering presentations at inter-university alliance meetings, Coordinated sponsorship acquisition and fund management involving corporate partners and university foundations
- 2019 (2nd Place Winner): Proposed and modeled joint design strategies as part of the design team, Conducted material studies and contributed to their physical implementation

Publication

EasyShade: Generating a Custom Lampshade with Stereographic-pattern Projection

Jul. 2024

Co-Author

- Demo in 2024 ACM Symposium on Computational Fabrication
- Fully engineered an end-to-end, “one-click” lampshade generation algorithm that automates the process from pattern projection to fabrication-ready outputs, generating precise laser-cutting vector drawings for joinery and 3D-printable geometry from a single input image.
- The input image is converted into vector data, traced, and re-mapped through a stereographic projection process to reconstruct the original image in a continuous geometric system.

Awards & Honors

Selected Awards

Second Prize, Korea Student Architecture Awards	Oct. 2022
Second Prize, Hanyang University Architecture Graduation Exhibition	Jul. 2022
Selected Proposal, Seoul 100-Year Masterplan Exhibition Open Call	Sep. 2023
2nd Place Winner, AI Heritage Competition, Gongju City Government	May 2024

Professional Competition Awards (as part of JK-AR)

4th Prize, International Architecture Competition For Gyeongju Integrated Cultural Library	Dec. 2025
3rd Prize, 10-27 Memorial Museum Architectural Design Competition	Apr. 2024
3rd Prize, Sangmu Incineration Plant Regeneration Project Competition	Aug. 2022

Awards

Honorable Mention, Ulsan Public Design Competition	Oct. 2024
Grand Prize, Seodaemun-gu Temporary Fence Design Competition	Jun. 2023
Honorable Mention, “Seoul Is Museum” Public Art Artist Discovery Competition	Aug. 2023
Grand Prize, Dongjak-gu Temporary Fence Design Competition	Dec. 2023
Honorable Mention, LAGI 2022 Mannheim, Land Art Generator	Nov. 2022
Honorable Mention, Seocho Smart City Living Lab Competition	Oct. 2022
Honorable Mention, Jeonglim Student Architecture Awards	Mar. 2022
Honorable Mention, 24th Gyeonggi Architecture Culture Awards	Aug. 2019
2nd Prize, 8th UAUS–Seoul Market 21	Jun. 2019
Honorable Mention Low-Utilization Urban Space Innovation Competition	Apr. 2019
2nd Prize, 7th UAUS–Seoul Forest Project	Jun. 2018

Exhibition

2024 Digital Heritage Festa Exhibition, Art Center GOMA, Gongju, South Korea	May. 2024
Land, Infrastructure and Transport Technology (LITT) Fair 2023, COEX, Seoul, South Korea	Apr. 2023
2022 Student Architectural Project Exhibition	Oct. 2022
2019 Seoul Biennale of Architecture and Urbanism, Live Project Exhibition, Seoul, South Korea	Sep. 2019
2019 UAUS Pavilion Exhibition, Seoul Market 21, Trukseom Park, Seoul, South Korea	Jun. 2019
Design Exhibition of Seoul-Type Urban Space Innovation Idea, Seoul, South Korea	Apr. 2019
2018 UAUS Pavilion Exhibition, Seoul Forest:Planting Joy, Seoul Forest Park, Seoul, South Korea	Jun. 2018

Academic Honors & Scholarships

Graduated Cum Laude, Hanyang University	Feb. 2023
Merit-Based Scholarship, Hanyang University	Fall Semester, 2019

Volunteer Works & Leaderships

Mandatory Military Service, Engineer Battalion, 28th Infantry Division	Nov. 2019 - Jun. 2021
President of Architecture Student Council	Mar. 2018 - Feb. 2019
Volunteer for Rural Communities during Summer Vacation	Jul. 2018
Architectural Workshop with High School Students	Jul. 2017

Skills

3D Software: Rhinoceros 3D + Grasshopper / SketchUp / Blender
Visualization Tools: Stable Diffusion-Comfy UI / V-ray / Lumion / Enscape
Editing Tools: Adobe Photoshop / Illustrator / InDesign / AutoCAD
Programming Language: Python